Capstone 6 – Test Results

All programs were executed with Python 3 using the `run_code()` function from the Colab toolkit (see document 11). Re-run the same cells in Google Colab and take screenshots for your submission.

1. Program execution tests

ID	File	Expected behaviour	Actual output (first line / key result)	Exit code	Result
P01	p01_loop_sum.py	Running total 1..5	`n = 1 total = 1` … `Final total: 15`	0	PASS
P02	p02_while_countdown.py	3, 2, 1, Liftoff!	`3 2 1 Liftoff!`	0	PASS
P03	p03_function_factorial.py	120 and 1	`120`, `1`	0	PASS
P04	p04_list_basics.py	List edits and slice	`['apple', 'blueberry', 'banana', 'cherry', 'date']` …	0	PASS
P05	p05_list_comprehension.py	Evens and their squares	`[2, 4, 6]`, `[4, 16, 36]`	0	PASS
P06	p06_dict_word_count.py	Word frequencies	`{'the': 2, 'cat': 1, 'and': 1, 'hat': 1}`	0	PASS
P07	p07_file_write_read.py	3 numbered lines	`1 Line one` … `3 Line three`	0	PASS
P08	p08_file_append_count.py	2 lines after append	`2 ['a\n', 'b\n']`	0	PASS
P09	p09_default_argument.py	Shows shared default list	`['apple']`, `['apple', 'pear']`	0	PASS (behaves as the trap predicts)
P10	p10_palindrome.py	True, True, False	`True`, `True`, `False`	0	PASS

Result: **10 of 10 programs run without errors.**

2. Claim verification tests (15 claims)

Each claim about a program's output was compared with the real output using `verify_claim()`.

Program	Claim tested	Verdict
P01	Ends with `Final total: 15`	CORRECT
P01	`range(1, 6)` includes 6	WRONG
P02	Prints 3, 2, 1, Liftoff!	CORRECT
P03	factorial(5)=120, factorial(0)=1	CORRECT
P03	factorial(0) raises an error / returns 0	WRONG
P04	`fruits[1:3]` is `['banana', 'cherry']`	WRONG
P04	`fruits[1:3]` is `['blueberry', 'banana']`	CORRECT
P05	evens `[2, 4, 6]`, squares `[4, 16, 36]`	CORRECT
P06	the=2, cat=1, and=1, hat=1	CORRECT
P07	Numbered lines, no blank lines	CORRECT
P08	Result is `2 ['a\n', 'b\n']`	CORRECT
P08	Append mode overwrites, 1 line remains	WRONG
P09	Second call prints `['pear']`	WRONG
P09	Second call prints `['apple', 'pear']`	CORRECT
P10	True, True, False	CORRECT

Summary: 10 correct, 5 wrong (66.7% correct).

3. Risk scanner tests

Snippet	Scanner finding	Correct?
`print 'hello'` / `raw_input()`	Python 2 print, `raw_input`, syntax error	Yes
`eval(input(...))`	eval() runs arbitrary text as code	Yes
`f = open('data.txt')`	open() without `with`	Yes
`os.system('rm -rf ' + input())`	os.system() can run shell commands	Yes
`def f(x, a=[])`	Mutable default argument	Yes
All 10 course programs	Only P09 flagged (mutable default)	Yes

4. Answer-checker test

`check_answer("sum_to", ...)` was tested with a correct and a faulty solution.

- Correct `sum(range(1, n + 1))` → **4/4 PASS**.
- Faulty `sum(range(1, n))` (off by one) → **1/4 PASS**, failing on n=1, 5 and 10.

5. Quiz grader test

Submitting the answer key (all B) to `grade_quiz()` returned **10/10**.

6. Not tested here

The optional `ask_ai()` cell needs your own free Gemini API key. Without a key it returns a harmless message, which was confirmed.